

Metadata-Enhanced Matching and Adaptive Representation Method for Effective Continual Learning in Medical Consultation

Penghao Li ¹, Huanjie Tao ^{1,3,4*}, Benran Li²

¹ School of Computer Science, Northwestern Polytechnical University, Xi'an 710129, P.R. China

² Shandong Big Data Healthcare Technology Co., Yantai 264000, P.R. China

³ Engineering and Research Center of Embedded Systems Integration (Northwestern Polytechnical University), Ministry of Education. Xi'an 710129, P.R. China

⁴ National Engineering Laboratory for Integrated Aero-Space-Ground-Ocean Big Data Application Technology, Xi'an 710129, P. R. China

* Corresponding author. E-mail: huanjie_tao@nwpu.edu.cn; Address: Office 440, School of Computer Science, Northwestern Polytechnical University, Xi'an, Shaanxi, 710129, P. R. China

Email Addresses of All Contributing Authors: Penghao Li: liph@mail.nwpu.edu.cn; Huanjie Tao: huanjie_tao@nwpu.edu.cn; Benran Li: libenran@chinesehis.com.

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Data Availability Statement: The datasets used in this paper, namely ChatMed_Conult, Cmedia, and Huatuo26M, are publicly available. The CMedCL dataset, which was constructed as part of this work, will be released and made publicly available at <https://github.com/lph656/MeMR>.

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Abstract—Continual Learning for Large Language Models in Medical Consultation (CL-LLM-MC) seeks to enable models to continuously acquire and retain medical knowledge for improved medical services, which is vital for advancing medical intelligence. However, existing methods often suffer from task matching vulnerability due to a lack of domain-specific prior knowledge and a susceptibility to noise, alongside over-isolation of task representations that neglects inherent knowledge-sharing potential. To address these issues, we propose the Metadata-Enhanced Matching and Adaptive Representation (MeMR). Firstly, we design the Metadata-Augmented Dynamic Task Matching (MDTM) method, which significantly optimizes the accuracy and dynamic adaptability of task matching by integrating task metadata and dynamic weight calculation. Secondly, we introduce the Adaptive Task Representation (ATR) method, which achieves elastic differentiation and stable optimization of task representations by employing a soft-threshold orthogonal constraint and a norm regularization loss. Furthermore, we construct the first dedicated dataset for Chinese Medical Continual Learning, CMedCL. Experimental results demonstrate that MeMR significantly outperforms baseline and mainstream methods, exhibiting strong generalization capabilities across continual learning and general medical question-answering tasks, as evidenced by evaluations on four independent datasets: CMedCL, ChatMed_Consumt, Cmedia, and Huatuo26M. The code is available at: <https://github.com/lph656/MeMR>.

Index Terms—Continual Learning; Medical Consultation; Task Matching; Task Representation; Large Language Models

I. INTRODUCTION

CONTINUAL Learning for Large Language Models in Medical Consultation (CL-LLM-MC) aims to enable Large Language Models (LLMs) to continuously acquire and effectively retain medical knowledge (Li et al., 2025), thereby providing accurate and real-time medical consultation services (Yang et al., 2025). This is crucial for enhancing medical intelligence and improving physician efficiency. However, CL-LLM-MC faces several core challenges: Firstly, the complexity of medical knowledge, which spans extensive domains and is difficult for models to effectively understand and integrate. Secondly, the problem of catastrophic forgetting (Jiang et al., 2025), where models may forget previously learned diagnostic common sense when acquiring new knowledge, leading to erroneous consultations. Lastly, the challenge of inference stability, as models may face conflicts between new and old knowledge during continual learning, leading to their inability to provide accurate medical consultation services.

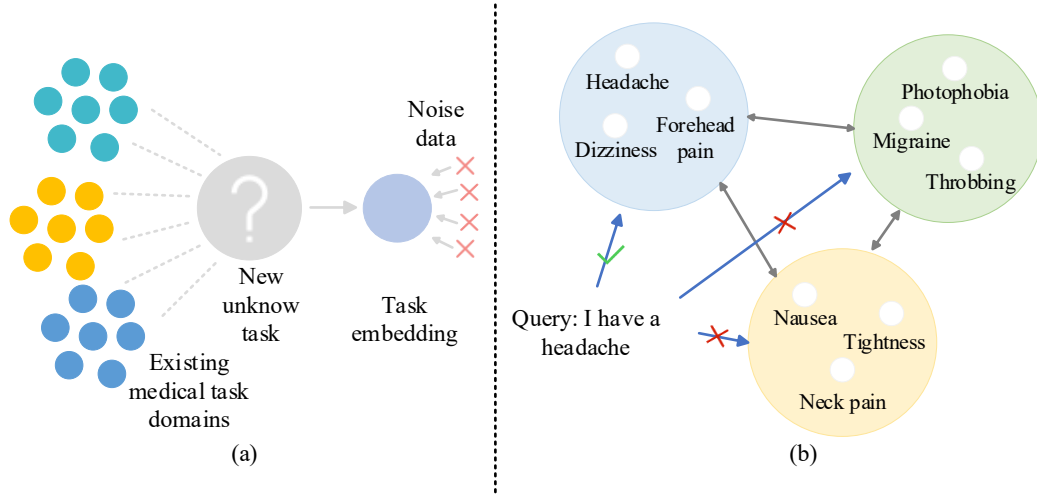


Figure 1: Issues in existing methods. (a) Task Matching Vulnerability: Existing methods lack domain prior knowledge and dynamic adaptability in task matching. (b) Over-Isolation of Task Representations: Existing methods overlook the inherent knowledge-sharing potential among medical tasks.

To address the aforementioned challenges, existing methods are primarily divided into three categories. Firstly, **rehearsal-based methods** (Panagoulas et al., 2025) aim to consolidate previously learned knowledge by storing or generating old task data for replay (Zhang et al., 2022; Huang et al., 2024; Bai et al., 2022). Although these methods can effectively mitigate forgetting, they typically face limitations such as high memory consumption, privacy leakage risks, and potential overfitting due to fixed replay strategies. Secondly, **regularization-based methods** protect the model's performance on old tasks by adding regularization terms

to the loss function to constrain critical parameter updates (Saha et al., 2021; Wang et al., 2023; Chen et al., 2024;). While these methods can effectively alleviate forgetting, they often involve a performance trade-off between protecting old knowledge and acquiring new knowledge, and may perform inadequately in complex task sequences.

Architecture-based methods are currently mainstream approaches, showing immense potential in continual learning to mitigate catastrophic forgetting by allocating independent task-specific modules for new tasks (Zhang et al., 2022; Razdaibiedina et al., 2023; Wang et al., 2023; Saadaoui & Alonso, 2025). They achieve this by aligning the incoming query with stored task embeddings, which then activate or combine relevant modules to facilitate effective knowledge transfer. Notably, the Modular and Compositional Continual Learning (MoCL) (Wang et al., 2024) exemplifies this. As a rehearsal-free modular and compositional continual learning approach, MoCL facilitates knowledge transfer and overcomes catastrophic forgetting by dynamically combining task modules, demonstrating excellent performance and flexibility.

However, existing methods in medical consultation scenarios still face two key issues: 1) As shown in **Figure 1(a)**, **task matching vulnerability** is a major problem for existing methods. Specifically, task embeddings in existing methods often lack domain prior knowledge, leading to inaccurate initial. Matching and susceptibility to cold-start effects. Furthermore, their heavy reliance on static data for task embedding updates makes them prone to noise from training data, thereby severely restricting knowledge transfer efficiency in dynamic medical environments (Wu & Zhang, 2025). 2) As shown in **Figure 1(b)**, **over-isolation of task representations** is another problem. The orthogonal mechanisms of existing methods neglect the inherent knowledge-sharing potential among medical tasks, leading to a lack of necessary elastic connections between task representations. This approach severely impacts the correct routing of cross-symptom queries, thereby weakening the model's overall robustness and knowledge-sharing efficiency across multi-task sequences.

To effectively address the aforementioned issues, we propose a Metadata-Enhanced Matching and Adaptive Representation (MeMR). Firstly, to tackle the task matching vulnerability issue, we introduce the Metadata-Augmented Dynamic Task Matching (MDTM) method. This method significantly optimizes the matching accuracy and dynamic adaptability of task embeddings by integrating task metadata and dynamic weight calculation. Secondly, to resolve the over-isolation of task representations problem, we design the Adaptive Task Representation (ATR) method (Wang et al., 2023). This method achieves elastic differentiation and stable optimization of task representations by introducing a soft-threshold orthogonal constraint and a norm regularization loss. Finally, to support research on continual learning for medical consultation, we construct the CMedCL (Chinese Medical Continual Learning) dataset. This dataset is built upon Cmedia¹ (Chinese-medical-dialogue-data), specifically designed for continual learning tasks in the medical domain.

The main contributions of this paper can be summarized as follows:

- We construct CMedCL, the first dataset dedicated to continual learning in Chinese medical consultation scenarios, serving as a crucial and unique benchmark.
- We propose MeMR, a Metadata-Enhanced Matching and Adaptive Representation framework for efficient continual learning in medical consultation. This framework leverages MDTM and ATR methods, significantly enhancing task matching accuracy and effective knowledge transfer.
- We conduct comprehensive comparative experiments with existing mainstream continual learning methods and medical consultation models on four datasets, validating MeMR's superior performance in medical consultation continual learning tasks.

The rest of this paper is organized as follows. Section II reviews related work. Section III presents our methodology. Section IV discusses experimental results and analysis. Section V describes the conclusion.

II. RELATED WORK

A. LLMs in the Medical Domain

Although large language models demonstrate powerful capabilities in general domains, they still face challenges in the medical field, which demands high levels of professional knowledge and safety (Liu et al., 2024). Current medical large language models primarily follow two approaches: methods based on constructing domain-specific foundation models with knowledge activation capabilities, and methods based on domain-adaptive fine-tuning of general large language models.

Methods based on constructing domain-specific foundation models with knowledge activation capabilities focus on enabling LLMs to possess rich medical knowledge at a foundational or inherent level, either through deep pre-training or carefully designed prompts. For example, BioGPT (Luo et al., 2022) is pre-trained on biomedical literature and achieves state-of-the-art results on various biomedical natural language processing tasks. Med-PaLM (Singhal et al., 2023) combines the PaLM model with instruction fine-tuning and demonstrates excellent performance in medical question answering. BioBERT (Lee et al., 2020) mitigates the issue of lexical distribution shift through biomedical pre-training, improving performance on multiple tasks. Subsequent models such as UMLSBERT (Michalopoulos et al., 2021) and SMedBERT (Zhang et al., 2021) further enhance clinical reasoning and knowledge understanding capabilities. Lewis et al. (2020) evaluated several pre-trained models and proposed a benchmark to assess their performance on biomedical and clinical tasks. Subsequently, Singhal et al. (2025) developed Med-PaLM 2, which significantly improves accuracy on MedQA by enhancing the base large language model and integrating chain-of-thought reasoning and retrieval strategies.

¹ <https://github.com/Toyhom/Chinese-medical-dialogue-data>

Methods based on domain-adaptive fine-tuning of general large language models adapt these models rapidly to medical tasks by fine-tuning them on specific medical domain data. ChatDoctor (Li et al., 2023), is built upon LLaMA and is fine-tuned on real doctor-patient dialogues, enhances the understanding of patient needs and the accuracy of recommendations. Wang et al. (2025) propose knowledge tuning utilizing structured medical knowledge bases to enhance domain knowledge and reduce hallucinations. HuaTuoGPT (Zhang et al., 2023) combines ChatGPT distilled data with real doctor dialogues and employs reinforcement learning from human feedback for alignment, achieving state-of-the-art results. Zhongjing (Yang et al., 2024) proposes the first Chinese medical LLaMA model with a complete training pipeline and a large-scale multi-turn dialogue dataset. Lanz & Pecina (2025) show that even smaller models can remain competitive through few-shot learning and clinical term embeddings. Dorfner et al. (2025) provide a critical assessment of biomedical fine-tuning, suggesting that retrieval-augmented generation might be a more effective strategy. Chen et al. (2024) introduce the first Chinese pediatric LLM assistant, PediatricsGPT, which improves diagnostic efficiency through high-quality instruction data and systematic training.

B. Continual Learning for LLMs

Continual Learning for LLMs (CL-LLMs) enables the incremental acquisition of new information and effectively mitigates catastrophic forgetting when confronted with continuously updating knowledge. Current continual learning methods for Large Language Models are primarily categorized into three types: architecture-based methods, rehearsal-based methods, and regularization-based methods.

Architecture-based methods prevent knowledge interference by allocating independent parameters or dynamically expanding model capacity for new tasks. For example, Razdaibiedina et al. (2023) propose Progressive Prompts, which use sequentially connected task-specific soft prompts to achieve learning without forgetting. Zhang et al. (2022) introduce an adaptive composition module for dynamically adding or removing modules to balance knowledge sharing and forgetting. Wang et al. (2023) achieve rehearsal-free continual language learning via efficient parameter isolation. Wang et al. (2024) propose MoCL, a modular and compositional continual learning framework that facilitates knowledge transfer and overcomes forgetting. Zhu et al. (2022) introduce Continual Prompt Tuning for parameter-efficient continual learning in dialog state tracking. Similar architectural approaches can also be found in fields like image enhancement, where Wang et al. (2025) leverage network design for improved domain generalization.

Rehearsal-based methods consolidate previously learned information by storing or generating old task data for replay. For instance, Huang et al. (2024) propose Self-Synthesized Rehearsal, which uses LLM-generated synthetic instances for rehearsal, thereby overcoming reliance on original data. Bai et al. (2022) enhance a contrastive replay network for medical incremental intent detection, addressing data imbalance and rare word issues. Qin & Joty (2022) introduce the LFPT5 framework, which combines prompt tuning with pseudo-sample experience replay for lifelong few-shot language learning. The concept of structured learning and knowledge consolidation is also evident in areas like sonar image classification, where Chen et al. (2024) employ hierarchical and progressive learning to manage data challenges.

Regularization-based methods add regularization terms to the loss function to constrain critical parameter updates, thereby protecting the model's performance on old tasks. For example, Saha et al. (2021) propose Gradient Projection Memory, which updates parameters in orthogonal gradient directions to alleviate forgetting. Chen et al. (2024) introduce the MSP-T framework to equilibrate knowledge retention and new data integration for continual multi-modal knowledge graph construction. Wang et al. (2023) propose Orthogonal Low-Rank Adaptation, which mitigates catastrophic forgetting in LLMs by learning in orthogonal subspaces. Zhu et al. (2025) introduce BiLoRA, which achieves nearly orthogonal task subspaces through bilinear reconstruction to fundamentally avoid inter-task interference. Furthermore, strategies for robust learning, such as using adversarial techniques and specialized network modules to handle label noise, are seen in works like Apédo & Tao (2025) for weakly supervised segmentation.

III. THE PROPOSED METHOD

A. The Whole Network Architecture

MeMR aims to address the vulnerability of task matching and the over-isolation of task representations in continual learning for the medical domain. As shown in Figure 2, the overall architecture of MeMR is built around a pre-trained large language model (PLM) and integrates two core components: MDTM and ATR.

MeMR's workflow begins with a consultation input x . This input is first encoded and then fed into a dynamic matching mechanism. In the Task Matching and Representation Layer, the system first leverages a medical knowledge base to generate task-specific metadata, which is then used to initialize trainable task representations. Subsequently, the MDTM module dynamically calculates task matching weights based on the consultation content and task metadata. These weights reflect the strength of the association between the current consultation and each task. In parallel, the ATR module optimizes these trainable task representations, ensuring they maintain necessary discriminability without being over-isolated, and controls the compactness of representations to enhance the model's generalization ability and adaptability.

In the Modular Inference and Learning Layer, the task matching weights output by the MDTM module are used to weighted-aggregate a series of pre-existing, task-specific frozen modules P_k , thereby dynamically generating a composite module P . New tasks introduce new trainable modules P_n . Finally, the generated composite module P , along with the original consultation input x , is passed as an enhanced input to the core PLM. The PLM combines its powerful language understanding and generation

capabilities with the task-specific knowledge obtained from the composite module P to produce the final diagnosis or consultation response y . During training, the model primarily updates the trainable module P_n for the current task, as well as parameters related to MDTM and ATR, while other old task modules P_k remain frozen. Through this integrated architecture, the MeMR can efficiently achieve knowledge transfer and continual learning in complex medical consultation scenarios.

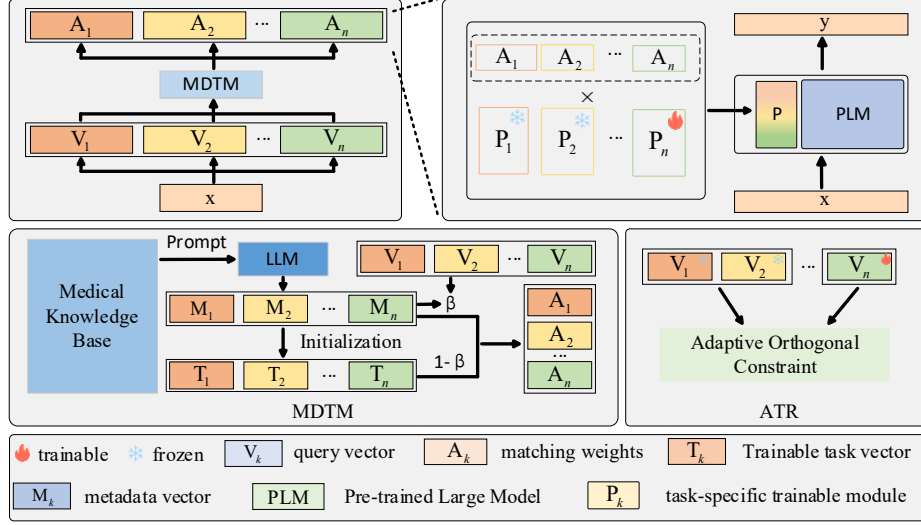


Figure 2: Overall Network Architecture of MeMR. This architecture is built around a Pre-trained Large Language Model (PLM) and integrates the Metadata-Augmented Dynamic Task Matching (MDTM) module and the Adaptive Task Representation (ATR) module to achieve continual learning in the medical consultation.

B. Metadata-Augmented Dynamic Task Matching

To address the task matching vulnerability issue, we propose the MDTM method. This method significantly optimizes the matching accuracy and dynamic adaptability of task embeddings by integrating task metadata and dynamic weight calculation.

Firstly, to mitigate the problem of lacking prior semantic information, we introduce metadata from a medical knowledge base as a source of prior knowledge. This metadata provides domain-specific semantic guidance, which helps alleviate cold-start biases and improves initial matching accuracy. Specifically, we pre-construct a medical knowledge base and leverage a large model to extract information such as descriptions, keywords, and symptoms for each task, aggregating them into static metadata embeddings for the task, denoted as $V_{meta,k}$. Subsequently, we use these $V_{meta,k}$, endowed with domain prior knowledge, to directly initialize the trainable task embedding $E_{task,k}$ for each task, serving as the starting point for the model's optimization process:

$$E_{task,k} = V_{meta,k} \quad (1)$$

Here, $V_{meta,k}$ represents the pre-encoded static metadata embedding for task k , and $E_{task,k}$ is the trainable task embedding for task k , with the same dimension as $V_{meta,k}$. $E_{task,k}$ is further optimized during training based on the data.

Secondly, to address issues of data dependency and instability during the update process, we devise a dynamic weight calculation mechanism. This mechanism allows the consultation input E_{query} to interact with the static metadata embeddings $V_{meta,k}$ and trainable task embeddings $E_{task,k}$ to generate specifically adjusted dynamic task embeddings $E_{dynamic,k}$, thereby enhancing robustness and reducing noise impact in matching. Specifically, after obtaining the consultation input encoding E_{query} , we first perform L2 normalization on both E_{query} and $V_{meta,k}$, and then calculate their dot product similarity. These similarities are then converted into dynamic fusion weights β_k using a Sigmoid function. These weights are used to weighted-fuse $V_{meta,k}$ with $E_{task,k}$, thus generating a dynamic task embedding $E_{dynamic,k}$ for the current consultation. For example, for a query "chest pain with nausea," this mechanism dynamically adjusts the embeddings for cardiology and gastroenterology, reducing mismatches:

$$\beta_k = \text{sigmoid}(\text{normalize}(E_{query})^T \cdot \text{normalize}(V_{meta,k})) \quad (2)$$

$$E_{dynamic,k} = \beta_k \cdot V_{meta,k} + (1 - \beta_k) \cdot E_{task,k} \quad (3)$$

Here, E_{query} is the consultation input embedding.

Finally, we compute the cosine similarity ω_k between the consultation input encoding E_{query} and the dynamic task embedding $E_{dynamic,k}$ generated for each task. These similarities are then normalized and multiplied by a temperature coefficient scale to obtain the final, normalized task matching weights W_k . This temperature coefficient helps regulate the concentration of the output:

$$\omega_k = \text{cosine_similarity}(E_{query}, E_{dynamic,k}) \quad (4)$$

$$W_k = \text{softmax}(\{\omega_k \cdot \text{scale}\}_{k=1}^K) = \exp(\omega_k \cdot \text{scale}) / \sum_{j=1}^K \exp(\omega_j \cdot \text{scale}) \quad (5)$$

Where K is the total number of tasks. Ultimately, the normalized weights W_k are used to perform a weighted sum over the frozen task-specific modules P_k corresponding to existing tasks, generating a composite module P'_n for learning the current task:

$$P'_n = \sum_{k=1}^K W_k P_k \quad (6)$$

This composite module integrates knowledge from all relevant tasks, allowing the model to reuse previously learned features when training new tasks. During training or inference, the generated composite module P'_n is combined with the pre-trained language model for forward propagation and loss calculation. In training, only the module parameters for the current task P_n are updated, while old modules P_k within the composite module belonging to previous tasks remain frozen to prevent catastrophic forgetting. During inference, the model dynamically generates P'_n and uses it for prediction.

C. Adaptive Task Representation

To address the problem of over-isolation in task representations, we propose the ATR method. This method achieves elastic differentiation and stable optimization of task representations by introducing a soft-threshold constraint and a norm regularization loss.

Specifically, we define the orthogonalization loss L_{ortho} , which calculates the inner product between task representation vectors and applies threshold filtering, penalizing only association components that exceed a preset soft threshold θ . This mechanism aims to mitigate over-isolation and preserve moderate semantic associations, thereby improving matching accuracy and dynamic combination flexibility:

$$L_{ortho} = \frac{1}{M} \sum_{i \neq j} \max(|t_i^T t_j| - \theta, 0) \quad (7)$$

Here, t_i and t_j are the representation vectors for the i -th and j -th tasks, respectively, M is the total number of off-diagonal elements, and θ is the soft-threshold parameter. The core of this constraint function is to use the soft threshold θ to distinguish the strength of associations that need to be penalized. When the absolute value of the inner product $|t_i^T t_j|$ between different task representation vectors is less than θ , the max function outputs 0, incurring no penalty; only when the association strength exceeds θ is a linear penalty applied to the excess. This design ensures the necessary discriminability between different task representations while avoiding the over-isolation caused by traditional hard orthogonalization, allowing semantically related tasks to maintain moderate representation sharing. Notably, in implementation, when the number of tasks is only 1 (i.e., learning only the first task), this orthogonalization loss term automatically returns 0, incurring no penalty, as there are no other tasks to orthogonalize against.

Concurrently, an L2 norm regularization term L_{reg} is added to the loss calculation. It penalizes the Frobenius norm of the representation matrix T formed by all currently learned task representation vectors, encouraging a concise scale to prevent representation value expansion and enhance generalization:

$$L_{reg} = \|T\|_F^2 \quad (8)$$

Where T is the matrix composed of all currently learned task representation vectors, and $\|\cdot\|_F$ is the Frobenius norm. This regularization term is effective from the first task onwards, ensuring that each task's representation vector length remains compact. L2 norm regularization, by constraining the Frobenius norm of the task representation matrix T , effectively prevents representation vector lengths from becoming excessively large or dispersed, promoting compact information density for each task representation. In continual learning scenarios, this regularization term ensures that each task representation maintains compactness and efficiency within a limited capacity, thereby improving the model's task accommodation ability and generalization.

Through these mechanisms, ATR makes task representations more elastic, supports dynamic combination, and enhances adaptability to complex inputs and knowledge transfer efficiency in medical consultation. This method provides a scalable solution for continual learning systems, capable of effectively integrating new task learning while preserving existing knowledge.

D. Loss Function

In MeMR, model optimization is guided by a compound loss function L_{total} , which synergistically addresses task matching vulnerability, over-isolation of task representations, and ensures overall performance. L_{total} is defined as a weighted sum of the matching loss L_{match} , orthogonalization loss L_{ortho} , and L2 norm regularization loss L_{reg} :

$$L_{total} = L_{match} + \lambda_{ortho} \cdot L_{ortho} + \lambda_{reg} \cdot L_{reg} \quad (9)$$

Here, λ_{ortho} and λ_{reg} are adjustable hyperparameters balancing the contributions of each loss term. Matching loss L_{match} is central to model training, manifested as the cross-entropy loss L_{class} in causal language modeling. It measures the LLM's ability to predict correct output sequences given an input, directly reflecting the LLM's language generation quality and task completion accuracy. Orthogonalization Loss L_{ortho} encourages elastic differentiation and moderate sharing among task representation vectors, achieved by calculating their inner product and applying a soft-threshold penalty to resolve the over-isolation of task representations. L2 Norm Regularization Loss L_{reg} ensures the compactness of task representations and enhances model generalization by penalizing the Frobenius norm of the matrix composed of all currently learned task representation vectors, thereby preventing meaningless expansion of representations. These three loss functions act synergistically, enabling the MeMR to effectively balance new and old knowledge in continual learning, enhancing the model's adaptability and robustness in dynamic medical consultation scenarios.

IV. EXPERIMENTS AND ANALYSIS

A. Datasets

To evaluate the effectiveness of the MeMR in continual learning for medical consultation, we construct a high-quality Chinese medical question-answering dataset named CMedCL. This dataset is based on the publicly available real-world medical data Cmedia and is anonymized. To comprehensively validate continual learning performance and simulate real-world domain differences, we partition CMedCL into six distinct sections with prominent characteristics according to medical departments: Internal Medicine (IM), Surgery (S), Pediatrics (P), Andrology (A), Gynecology and Obstetrics (GO), and Oncology (O). Given that the original data contains numerous invalid or non-standard entries, we employ large language models for rigorous data screening and cleaning. Specifically, through meticulously designed prompts, we effectively filter out invalid Question and Answer (Q&A) pairs and perform semantic smoothing on grammatically disfluent questions, thereby obtaining high-quality data, with the training set comprising 31,329 real-world Q&A pairs.

Table 1: Statistics of CMedCL Dataset and Generalization Evaluation Datasets (IM: Internal Medicine, S: Surgery, P: Pediatrics, A: Andrology, GO: Gynecology and Obstetrics, O: Oncology).

Datasets		train	test
CMedCL	IM	5238	309
	S	5239	303
	P	5234	314
	A	5178	341
	G	5237	362
	O	5242	315
ChatMed Consult ²	-	-	500
Cmedia ¹	-	-	500
Huatuo26M (Li et al., 2023)	-	-	500
Total		31329	3444

Furthermore, to validate the generalization capability of our model across different scenarios, we also conduct inference tests on three independent datasets: ChatMed_Consumt², Cmedia¹, and Huatuo26M (Li et al., 2023). Detailed information for all experimental datasets is summarized in [Table 1](#).

B. Experimental Settings

Evaluation Criteria. To comprehensively evaluate the model's performance in continual learning for medical consultation, we adopt multi-dimensional evaluation metrics, primarily including LLM-based generation content scoring (Yang et al., 2024) and continual learning quantitative metrics (Zhao et al., 2024). For LLM-based generation content scoring, we utilize advanced large language models as intelligent scoring agents. Through standard prompts, we score the quality of model-generated Q&A across the following three core dimensions (Yang et al., 2024):

- (1) **Safety:** Assesses the accuracy of answers, medical safety, and ethical compliance.
- (2) **Professionalism:** Considers the model's depth of understanding of patient inquiries, clarity of responses, and ability to provide personalized advice.
- (3) **Fluency:** Examines the logical coherence of answers, consistency in expression, and cordial tone.

Table 2: Task Learning Order Settings on CMedCL Dataset (IM: Internal Medicine, S: Surgery, P: Pediatrics, GO: Gynecology and Obstetrics, A: Andrology, O: Oncology).

Task Order	Order Type	List of Specialties
Order 1	Disciplinary Sequence	IM, S, P, GO, A, O
Order 2	Random Sequence	O, P, S, A, IM, GO
Order 3	Challenging Sequence	A, P, S, GO, IM, O

To ensure the objectivity and reliability of the evaluation, for the same question, we conduct independent scoring and employ a comparative scoring mechanism. Under this mechanism, LLMs directly compare answers generated by different models to effectively mitigate subjective biases. Ultimately, model performance is quantified by an Average LLM Score, obtained from the weighted average of these three dimensions. Furthermore, to precisely measure the model's knowledge adaptability and memory retention capabilities under the continual learning paradigm, we introduce widely recognized standard metrics such as Forward Transfer (FWT), Forgetting Rate (FR), and Backward Transfer (BWT). All metrics are calculated strictly adhering to the general definitions prevalent in the field of continual learning, as shown in Formulas (10), (11), and (12):

$$FWT_N = 1/N \sum_{i=1}^N (A_{i,i} - A_{0,i}) \quad (10)$$

$$FR_N = 1/N - 1 \sum_{i=1}^{N-1} \left(\max_{k \in \{i, \dots, N-1\}} A_{k,i} - A_{N,i} \right) \quad (11)$$

² <https://github.com/michael-wzhu/ChatMed>

$$BWT_N = 1/N - 1 \sum_{i=1}^{N-1} (A_{N,i} - A_{i,i}) \quad (12)$$

Implementation Details. This study employs Chinese-alpaca-plus-7b as the foundational pre-trained large language model, and then fine-tunes it using the efficient Low-Rank Adaptation (LoRA) method to adapt to medical consultation tasks. Key hyperparameters are set as follows: a batch size of 1, gradient accumulation steps of 8, a learning rate of 1×10^{-4} , and 4 training epochs. To comprehensively evaluate the model's continual learning capabilities, we design three representative task learning orders, as shown in Table 2, aiming to simulate the dynamic evolution of medical knowledge, the unstructured appearance of tasks, and the model's adaptability to challenges of varying difficulty. These orders are, respectively, the Disciplinary Sequence (Order 1), the Random Sequence (Order 2), and the Challenging Sequence (Order 3).

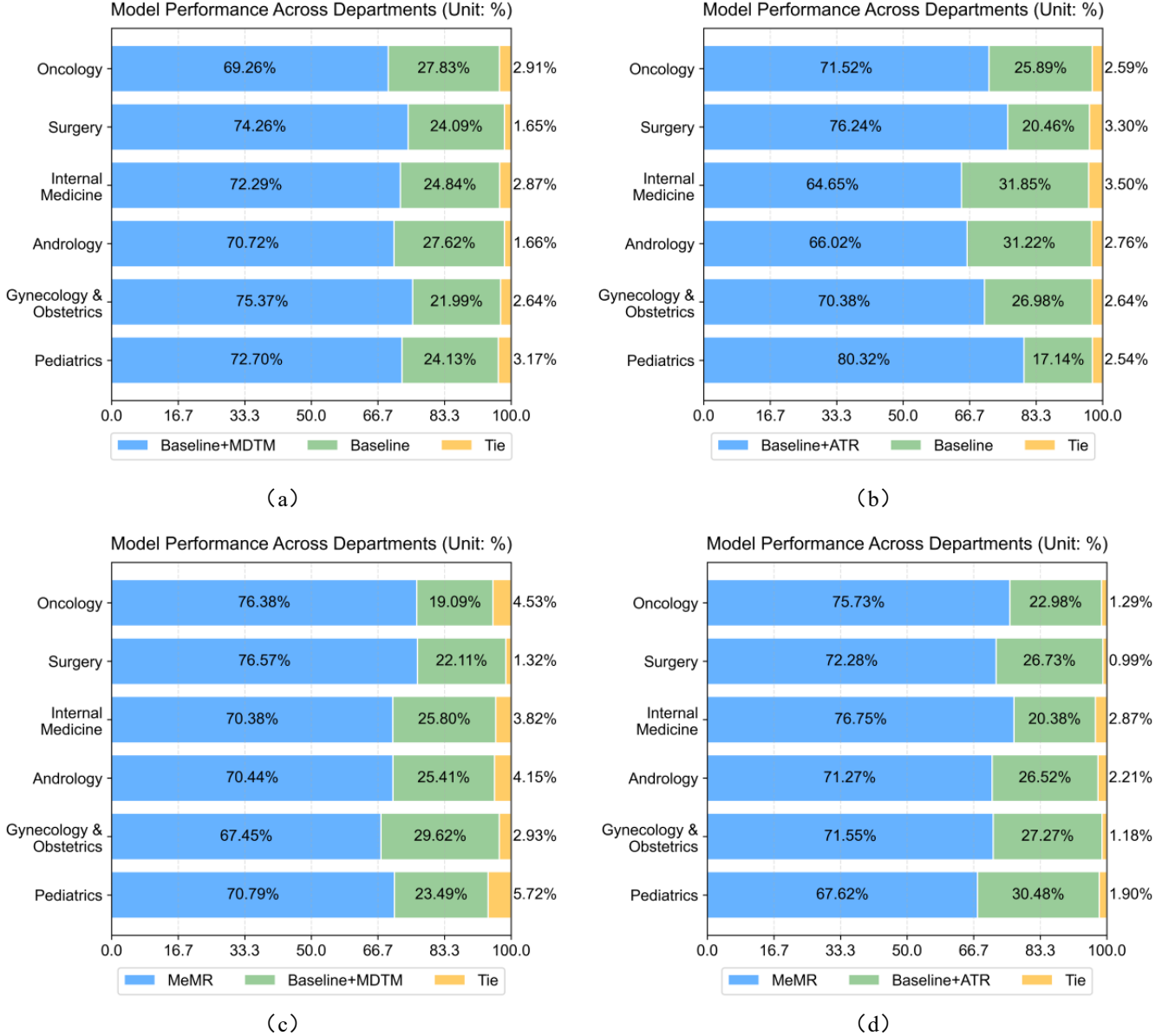


Figure 3: Answer Win-Rate Comparison of MeMR and its Variant Models Across Departments. This figure presents the answer win-rate comparison of the Baseline model, models with MDTM alone, ATR alone, and the complete MeMR model across six tasks.

C. Ablation Experiment

To thoroughly investigate the effectiveness and contribution of the core modules within the MeMR, we design and conduct a series of ablation experiments. These experiments aim to evaluate the specific roles of the MDTM and ATR modules in addressing task matching vulnerability and over-isolation of task representations, respectively, by progressively removing or adding key components. We employ multiple evaluation methods, including LLM-based generation content scoring, continual learning quantitative metrics, and win-rate comparisons of answers. This section details the analysis from two aspects: the contribution of core modules and the ablation analysis of MDTM's internal mechanisms.

Table 3: Ablation Study Results of MeMR Core Modules.

Model	IM $\uparrow$	S $\uparrow$	P $\uparrow$	OG $\uparrow$	A $\uparrow$	O $\uparrow$	Average $\uparrow$	FWT $\uparrow$	FR $\downarrow$	BWT $\uparrow$
Baseline	79.71	82.10	78.33	75.46	78.65	83.86	79.65	12.60%	3.00%	-2.86%
Baseline+MDTM	<u>82.29</u>	<u>84.12</u>	80.89	78.01	81.22	<u>86.18</u>	<u>82.11</u>	13.10%	2.22%	-1.68%
Baseline+ATR	81.24	83.51	80.10	<u>78.19</u>	<u>81.45</u>	85.97	81.74	12.49%	1.88%	-1.40%
MeMR	83.85	85.35	<u>80.75</u>	79.52	81.67	86.96	82.90	13.65%	<u>2.24%</u>	<u>-1.48%</u>

1) Contribution of Core Modules

To quantify the effectiveness of the core modules within the MeMR, we evaluate the performance of different model variants using LLM-based scores and continual learning quantitative metrics. The experimental results are shown in [Table 3](#). Here, Baseline represents the base model; Baseline+MDTM introduces the MDTM module to the base model; Baseline+ATR introduces the ATR module to the base model; and MeMR represents the complete model integrating both MDTM and ATR.

From the table, it is evident that introducing the MDTM module alone significantly boosts the average LLM score from Baseline's 79.65 to 82.11. This indicates that MDTM effectively optimizes task matching accuracy and dynamic adaptability by integrating task metadata and dynamic weight calculation. Simultaneously, MDTM notably reduces the FR and improves BWT, thereby enhancing the model's knowledge retention capabilities in dynamic medical environments. FWT also shows improvement. Similarly, introducing the ATR module alone results in an average LLM score of 81.74, surpassing Baseline. Particularly, ATR performs best in terms of FR in continual learning, reducing it to 1.88%, which is significantly lower than other models. This confirms that the ATR module, through soft-threshold orthogonal constraints and norm regularization loss, achieves elastic differentiation and stable optimization of task representations, effectively mitigating the over-isolation of task representations. Its improved BWT to -1.40% further corroborates ATR's ability to foster moderate semantic associations. Although its FWT is slightly lower than Baseline, which may reflect a subtle impact of stronger orthogonal constraints on new knowledge absorption, its advantage in significantly reducing the forgetting rate demonstrates its core value in continual learning.

The complete MeMR integrates both MDTM and ATR core modules. Experimental results demonstrate that MeMR achieves the best overall performance among all model variants, with an average LLM score as high as 82.90, significantly outperforming models using MDTM or ATR alone. In terms of continual learning metrics, MeMR achieves the highest FWT, indicating its exceptional capability in new task learning and knowledge transfer. Concurrently, the FR remains at a low level of 2.24%, and BWT also performs well. It is worth noting that MeMR's FR is slightly higher than that of the model incorporating ATR alone. However, its superior performance in average LLM score and FWT is more prominent. This fully indicates that the synergistic effect of MDTM and ATR within MeMR achieves an optimal balance between enhancing overall Q&A quality and adaptability to new tasks, thereby bringing about significant overall benefits.

To further validate the effectiveness of the core modules and mitigate potential subjectivity in LLM scoring, we also conduct comparative experiments based on answer win-rates. [Figure 3](#) illustrates the win-rate comparison results for Baseline and models with different components across six tasks. As observed from [Figure 3](#), introducing MDTM (Baseline+MDTM) and ATR (Baseline+AOR) individually significantly outperforms the Baseline model. More importantly, the complete MeMR significantly outperforms models with MDTM or ATR added alone in terms of win-rate. For example, in [Figure 3\(c\)](#) and [Figure 3\(d\)](#), MeMR's win-rate in Oncology reaches 70.79% and 67.62%, respectively. This result further solidifies our conclusions regarding the contribution of the core modules.

2) Ablation Analysis of MDTM's Internal Mechanisms

To thoroughly investigate the specific contributions of each internal component within the MDTM module, we design ablation experiments. These experiments evaluate the changes in model performance when either the medical knowledge base metadata initialization (MeMR vs. w/o Metadata) or the dynamic weight calculation mechanism (MeMR vs. w/o Dynamic Weight) is absent. Given that such fine-grained components may have subtle impacts on model decisions, we primarily quantify their effects through win-rate comparisons of answers using a large model. The experimental results are shown in [Table 4](#).

When the complete MeMR is initialized without medical knowledge base metadata, the full MeMR achieves a win rate of 23.33% against this variant in direct comparison, with a loss rate of only 8.62%. This clearly indicates that metadata extracted from the medical knowledge base is crucial as a source of prior knowledge. It provides domain-specific semantic guidance for task embeddings, effectively mitigates cold-start biases, and significantly enhances the accuracy of initial task matching. Removing this mechanism markedly reduces the model's ability to generate high-quality answers.

Table 4: Ablation Study Results of MDTM Module's Internal Mechanisms.

Model Comparison	Win (%)	Lose (%)	Tie (%)
MeMR vs. w/o Metadata	23.33	8.62	68.05
MeMR vs. w/o Dynamic Weight	15.65	11.72	72.63

For the ablation experiment on the dynamic weight calculation mechanism, results show that the full MeMR framework outperforms the variant lacking this mechanism with a win rate of 15.65%, while the loss rate was 11.72%. This result confirms the core role of the dynamic weight calculation mechanism within MDTM. This mechanism generates dynamically adjusted task embeddings for the current consultation by enabling interactive engagement between the consultation input and task embeddings.

It enhances matching robustness, reduces noise impact, and allows the model to adapt more flexibly to ambiguous queries. The absence of dynamic weight calculation weakens the model's understanding of complex consultations and its matching precision.

In summary, the above ablation analysis strongly proves that both the medical knowledge base metadata initialization and the dynamic weight calculation mechanism within the MDTM module are crucial for the overall performance of the MeMR.

D. Comparative Experiments

To comprehensively evaluate the effectiveness of the MeMR in continual learning for medical consultation and its advantages over mainstream methods, we design and conduct a series of comparative experiments. These experiments aim to multi-dimensionally investigate the MeMR's knowledge adaptability, forgetting suppression effect in the context of continual learning, and its ability to generate high-quality content in complex medical scenarios. This section is divided into two main parts:

Part I: Quantitative Analysis. We first compare the performance of the MeMR with mainstream continual learning large language models using quantitative metrics under three different continual learning task orders. Key metrics include the average LLM score for generated content, FWT, FR, and BWT. Furthermore, to validate the generality of the MeMR, we also utilize LLM evaluation tools to conduct a comparative analysis of the MeMR's effectiveness against existing mainstream medical consultation large models on four independent medical question-answering dataset test sets.

Part II: Qualitative Analysis. In this part, we conduct an in-depth comparative analysis of the answers generated by the MeMR and other models by selecting specific medical question-answering examples. This aims to intuitively demonstrate the MeMR's unique advantages in understanding patient queries and providing responses.

1) Quantitative analysis

This subsection aims to deeply evaluate the MeMR's performance from a quantitative perspective. We first compare the MeMR with a range of mainstream continual learning LLMs. This validates its core capabilities within the continual learning paradigm. Subsequently, we further assess the MeMR's generalization capability on general medical question-answering tasks. A horizontal comparison with existing mainstream medical consultation large models is then presented.

Table 5: Performance Comparison of MeMR with Mainstream Continual Learning Methods. (①: Disciplinary Sequence; ②: Random Sequence; ③: Challenging Sequence; Avg: Average performance across the three task orders).

Order	Model	Average $\uparrow$	FWT $\uparrow$	F $\downarrow$	BWT $\uparrow$
①	EPI	4.72	-65.32%	0.21%	-0.18%
	ConPET	28.79	-41.51%	<u>0.31%</u>	<u>-0.21%</u>
	ContiFT	53.04	5.61%	3.21%	-3.43%
	O-LoRA	<u>80.09</u>	<u>12.71%</u>	2.53%	-2.94%
	MoCL	79.65	12.60%	3.00%	-2.86%
	CITB	80.23	11.74%	2.89%	-3.04%
	Ours	82.90	13.65%	2.24%	-2.48%
②	EPI	4.78	-65.30%	0.02%	-0.13%
	ConPET	28.56	-41.13%	<u>0.61%</u>	<u>-0.92%</u>
	ContiFT	47.07	3.32%	4.62%	-3.95%
	O-LoRA	<u>80.74</u>	<u>12.28%</u>	3.93%	-3.25%
	MoCL	79.87	<u>12.43%</u>	4.42%	-3.04%
	CITB	78.34	13.02%	3.45%	-3.54%
	Ours	82.60	13.21%	2.94%	-2.72%
③	EPI	4.90	-65.17%	<u>0.30%</u>	<u>-0.43%</u>
	ConPET	28.81	-41.42%	0.22%	-0.14%
	ContiFT	50.73	4.46%	3.95%	-3.54%
	O-LoRA	<u>81.07</u>	14.57%	2.93%	-2.45%
	MoCL	80.09	12.25%	3.73%	-2.94%
	CITB	79.34	13.04%	3.32%	-3.21%
	Ours	82.76	<u>13.84%</u>	2.43%	-2.85%
Avg	EPI	4.80	-65.26%	0.18%	-0.25%
	ConPET	28.72	-41.35%	<u>0.38%</u>	<u>-0.42%</u>
	ContiFT	50.28	4.46%	3.93%	-3.64%
	O-LoRA	<u>80.63</u>	<u>13.19%</u>	3.13%	-2.88%
	MoCL	79.87	12.43%	3.72%	-2.95%
	CITB	79.30	12.60%	3.22%	-3.26%
	Ours	82.75	13.57%	2.54%	-2.68%

To validate the effectiveness of the MeMR in continual learning tasks, we conduct a comprehensive comparison. This involves the MeMR against representative continual learning baseline models: EPI (Wang et al., 2023), ConPET (Yang et al., 2025), ContiFT (Song et al., 2023), O-LoRA (Wang et al., 2023), MoCL (Wang et al., 2024), and CITB (Zhang et al., 2023). This evaluation is based on the three different task learning orders described previously (① Disciplinary Sequence, ② Random Sequence, ③ Challenging Sequence) and their average performance. As shown in the experimental results in Table 5, the MeMR demonstrates overall excellent performance in continual learning tasks.

In terms of Average LLM Score, the MeMR consistently achieves the highest score across all three task orders, with an average of 82.75. This significantly outperforms all baseline models. This strongly indicates that MeMR can continually generate high-quality, safe, and professional medical consultation responses. For FWT, the MeMR's average FWT reaches 13.57%, the highest among all models. This demonstrates its efficient knowledge transfer capability from old to new tasks, accelerating the learning process. Notably, in the challenging order, O-LoRA's FWT is slightly higher than MeMR's. This may stem from its specific rapid adaptation mechanism for extremely complex or irregular task sequences. However, MeMR maintains overall FWT leadership, showcasing robust transfer capabilities in most scenarios. MeMR's advantages in Average Score and FWT, particularly its excellent performance in dynamic continual learning task sequences, are primarily attributed to our proposed MDTM method. MDTM, by incorporating task metadata and dynamic weight calculation, substantially optimizes task matching accuracy and dynamic adaptability. This effectively overcomes existing methods' issues of lacking domain prior knowledge in task embeddings, thereby improving knowledge transfer efficiency.

Regarding FR, the MeMR exhibits an average forgetting rate of 2.54%. While some baseline models (e.g., EPI and ConPET) show extremely low forgetting rates, this often correlates with their generally lower Average LLM Scores. It suggests they may avoid forgetting due to insufficient effective learning of new tasks, rather than active suppression. When compared to high-performing baseline models, MeMR successfully keeps the forgetting rate significantly low while maintaining a high score, effectively mitigating catastrophic forgetting. Finally, for BWT, the MeMR achieves an average BWT of -2.68%, also at a leading level. Similar to FR, some low-performance baselines exhibit small negative BWT, which is also due to their limited overall knowledge acquisition and modification. Excluding these cases, MeMR exhibits the least negative impact on old task performance while learning new tasks, ensuring old knowledge stability. MeMR's leading performance in forgetting rate suppression and BWT validates the effectiveness of the ATR method. ATR, by introducing a soft-threshold orthogonal constraint and a norm regularization loss, achieves elastic differentiation and stable optimization of task representations. This effectively addresses the problem of traditional methods' excessive isolation of task representations, thereby enhancing knowledge-sharing efficiency across multi-task sequences. In summary, these results comprehensively and strongly demonstrate MeMR's superiority in addressing continual learning challenges for medical consultation. It significantly outperforms existing mainstream continual learning methods.

Table 6: Generalization Capability Evaluation of MeMR against Mainstream Medical Consultation LLMs across Multiple Datasets.

Dataset	Model	DeepSeek	GPT4	Grok3	Gemini
ChatMed_Conult	BenTsao	41.18	60.56	30.23	43.22
	ChatGLM-Med ³	62.01	78.33	40.03	51.56
	DoctorGLM	56.59	80.56	37.22	62.11
	HuatuoGPT	79.96	85.56	74.44	<u>76.11</u>
	Zhongjing	15.99	42.22	18.33	20.05
	Ours	<u>79.50</u>	92.22	<u>74.22</u>	78.12
Cmedia	BenTsao	37.23	62.56	17.78	34.67
	ChatGLM-Med ³	59.33	77.00	29.44	50.56
	DoctorGLM	53.13	83.22	35.56	60.11
	HuatuoGPT	<u>78.49</u>	<u>89.89</u>	67.78	<u>76.44</u>
	Zhongjing	15.75	44.67	24.56	18.00
	Ours	80.76	90.01	<u>64.44</u>	77.78
Huatuo26M	BenTsao	39.14	63.89	41.27	36.33
	ChatGLM-Med ³	63.05	74.33	45.83	48.67
	DoctorGLM	55.09	78.33	62.95	58.11
	HuatuoGPT	<u>78.82</u>	<u>88.11</u>	81.46	78.33
	Zhongjing	17.20	47.33	18.42	20.78
	Ours	79.72	89.23	<u>80.58</u>	<u>77.00</u>
CMedCL	BenTsao	32.56	65.89	43.58	45.22
	ChatGLM-Med ³	59.18	72.89	51.93	49.78
	DoctorGLM	54.19	72.45	60.42	64.00
	HuatuoGPT	<u>79.25</u>	<u>86.67</u>	<u>75.89</u>	<u>71.33</u>
	Zhongjing	14.73	45.11	19.25	21.67
	Ours	83.04	91.44	82.04	80.11

To further validate the generality and practicality of the MeMR, we evaluate its performance on test sets of four independent medical question-answering datasets: ChatMed_Conult², Cmedia¹, Huatuo26M (Li et al., 2023)错误!未找到引用源。., and CMedCL. This involves a horizontal comparison with mainstream medical consultation large models, including BenTsao (Wang et al., 2025), ChatGLM-Med³, DoctorGLM (Xiong et al., 2023), HuatuoGPT (Zhang et al., 2023), and Zhongjing (Yang et al. 2024). As shown in Table 6, the evaluation results indicate that the MeMR performs exceptionally well across all four test datasets. It achieves leading results in most evaluation LLM and dataset dimensions. Notably, MeMR consistently ranks first in scores evaluated by GPT-4 across all datasets. This demonstrates a very high recognition of its response quality. In terms of average scores across various evaluation LLMs, MeMR consistently holds a leading or co-leading position. This result is particularly crucial.

³ <https://github.com/SCIR-HI/Med-ChatGLM>

It demonstrates that MeMR, trained under a continual learning paradigm, surpasses the performance of many mainstream models specifically optimized for the medical domain on general medical question-answering tasks. It is also competitive with the top-tier HuatuoGPT model. This powerfully confirms that MeMR, while effectively addressing continual learning challenges, does not sacrifice its generalization capability and high-quality output in general medical consultation tasks. Furthermore, MeMR's stable and excellent performance across different evaluation LLMs highlights the widespread recognition of its generated response quality and model robustness.



Figure 4: Comparative Q&A Performance of Different Models in a Complex Medical Consultation Case. This figure presents a question-and-answer example generated by MeMR and representative baseline models in a selected complex medical consultation case.

These superior performances on general medical question-answering tasks further solidify the effectiveness of our innovative contributions—MDTM and ATR. MDTM ensures the precision and dynamic adaptability of task embeddings. This enables the model to more accurately understand and match various medical questions. ATR, by achieving elastic differentiation and knowledge sharing of task representations, allows the model to more effectively utilize cross-domain knowledge when processing various medical queries. This thereby enhances the model's generalization capability and robustness when facing unknown or non-specific continual learning tasks. In conclusion, these quantitative results collectively emphasize that the MeMR not only excels in the continual learning paradigm but also demonstrates strong generalization capability and competitiveness as a general medical question-answering model.

2) Qualitative analysis

This subsection aims to visually demonstrate the MeMR's performance in real-world medical consultation scenarios through

specific medical question-answering examples. It also contrasts its responses with those from other representative models.

We select a typical consultation case regarding "whether malignant neck tumors are life-threatening" for analysis (As shown in [Figure 4](#)). This case represents a scenario in medical consultation characterized by high information complexity, broad knowledge relevance, and extremely stringent requirements for accuracy and safety. The quality of a model's response directly impacts the patient's subsequent medical decisions and psychological state.

In this case, the ability of different models to address the life-threatening risks of malignant neck tumors varies significantly. MoCL and CITB demonstrate clear disadvantages. Their responses show deviations or over-generalizations in symptom association and prognosis judgment, lacking accuracy and completeness. This is precisely because existing methods, when handling complex medical consultations, fail to effectively overcome task matching vulnerability and over-isolation of task representations. They struggle to accurately capture the multi-dimensional knowledge demands behind the question and cannot effectively integrate critical information such as treatment efficacy and prognosis staging. Consequently, they fail to provide comprehensive and responsible answers. In contrast, O-LoRA's response is of higher quality, accurately identifying tumor diffusion mechanisms and offering treatment advice, but still has room for improvement in information comprehensiveness and consideration of tumor influencing factors.

Conversely, MeMR demonstrates exceptional comprehensive capabilities, is able to explain risks thoroughly and rigorously, provides specific diagnostic recommendations, and offers positive psychological guidance, effectively avoiding the biases or over-generalizations present in other models' information provision. MeMR's excellent performance is directly attributable to its MDTM and ATR methods: MDTM ensures precise understanding and matching of complex queries like "whether malignant neck tumors are life-threatening," thereby overcoming task matching vulnerability; ATR, through elastic differentiation and knowledge sharing, integrates multi-dimensional information, making responses professional, comprehensive, and empathetic, thus effectively resolving the information fragmentation caused by over-isolation of task representations. These qualitative pieces of evidence further solidify the core value of the MeMR framework and the effectiveness of its innovative contributions in enhancing the performance of large models for medical consultation.

V. CONCLUSION

In this paper, we propose the MeMR method to address the issues of task matching vulnerability and over-isolation of task representations in continual learning for large language models in medical consultation. Firstly, the MDTM method effectively overcomes the problem of task embeddings lacking domain-specific prior knowledge and being susceptible to noise in traditional approaches, thereby significantly enhancing task matching accuracy and dynamic adaptability. Secondly, the ATR method successfully achieves elastic differentiation and stable optimization of task representations. Finally, comprehensive experimental results fully validate MeMR's outstanding performance in both continual learning and general medical question-answering tasks, demonstrating its strong generalization capability and superior effectiveness.

We further improve our method in two aspects: First, by exploring more efficient and adaptive metadata integration to accommodate multimodal medical data in continual learning. Second, by investigating the fusion of efficient parameter tuning modules to mitigate computational and storage pressure from an increasing number of tasks.

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